ELITE IGCSE ACADEMY

Private Past Paper Solutions

Jan 2020 P1HR Worked Solutions

Jan 2020 | P1HR

27 questions ▪ question image followed by worked solution

PREPARED BY

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TEACHER EDITION

Worked answers included

PRIVATE

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PAST PAPER QUESTION

Q#: 1 Marks: 3

TOPIC: **Ratio Problem Solving**

Jan 2020 P1HR - Jan 2020 | P1HR

1 Brendon, Asha and Julie share some money in the ratios 3 : 2 : 6
The **total** amount of money that Asha and Julie receive is \$36

Work out the amount of money that Brendon receives.

\$...

(Total for Question 1 is 3 marks)

PAST PAPER SOLUTION

Q#: 1 Marks: 3

TOPIC: **Ratio Problem Solving**

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WORKED SOLUTION

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Topic check. Ratio Problem Solving. The tag is correct.

Method

The ratio is

$$\text{Brendon} : \text{Asha} : \text{Julie} = 3 : 2 : 6$$

Asha and Julie together have

$$2 + 6 = 8 \text{ parts}$$

These 8 parts are worth \$36, so one part is

$$36 \div 8 = 4.50$$

Brendon has 3 parts:

$$3 \times 4.50 = 13.50$$

FINAL ANSWER

Brendon receives \$13.50.

PAST PAPER QUESTION

Q#: 2 Marks: 3

TOPIC: **Fractions**

Jan 2020 P1HR - Jan 2020 | P1HR

2 Show that $3\frac{1}{5} \times 2\frac{5}{8} = 8\frac{2}{5}$

(Total for Question 2 is 3 marks)

PAST PAPER SOLUTION

Q#: 2 Marks: 3

TOPIC: **Fractions**

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WORKED SOLUTION

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Topic check. Fractions. The tag is correct.

Method

Convert the mixed numbers to improper fractions:

$$3\frac{1}{5} = \frac{16}{5}$$

$$2\frac{5}{8} = \frac{21}{8}$$

Now multiply:

$$\frac{16}{5} \times \frac{21}{8} = \frac{2 \times 21}{5}$$

$$= \frac{42}{5} = 8\frac{2}{5}$$

FINAL ANSWER

$8\frac{2}{5}$

PAST PAPER QUESTION

Q#: 3 Marks: 8

TOPIC: **Rearranging Formulas**

Jan 2020 P1HR - Jan 2020 | P1HR

3 (a) Make a the subject of $d = g + 2ac$

....

(2)

(b) Factorise fully $9ef - 12f$

....

(2)

(c) Expand and simplify $(x + 2)(x - 5)$

....

(2)

(d) Simplify fully $\dfrac{n^4 \times n^7}{n^5}$

....

(2)

(Total for Question 3 is 8 marks)

PAST PAPER SOLUTION

Q#: 3 Marks: 8

TOPIC: **Rearranging Formulas**

Jan 2020 P1HR - Jan 2020 | P1HR

WORKED SOLUTION

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Topic check. Rearranging formulas. The tag is acceptable in this mixed algebra question.

Solution

(a)

$$d = g + 2ac$$

$$d - g = 2ac$$

$$a = \frac{d - g}{2c}$$

(b)

$$9ef - 12f = 3f(3e - 4)$$

(c)

$$(x + 2)(x - 5) = x^2 - 3x - 10$$

(d)

$$\frac{n^4 \times n^7}{n^5} = n^{4+7-5} = n^6$$

FINAL ANSWER

$a = \frac{d-g}{2c}$, $3f(3e - 4)$, $x^2 - 3x - 10$, n^6

PAST PAPER QUESTION Q#: 4 Marks: 3

TOPIC: **Set Notation & Venn Diagrams**

Jan 2020 P1HR - Jan 2020 | P1HR

4 $B = \{b, l, u, e\}$

$G = \{g, r, e, y\}$

$W = \{w, h, i, t, e\}$

(a) List all the members of the set

(i) $B \cup G$

....

(ii) $W \cap G'$

....

(2)

Serena writes down the statement $B \cap G \cap W = \varnothing$

(b) Is Serena's statement correct?
You must give a reason for your answer.

....

....

....

(1)

(Total for Question 4 is 3 marks)

PAST PAPER SOLUTION

Q#: 4 Marks: 3

TOPIC: **Set Notation & Venn Diagrams**

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Topic check. Set notation and Venn diagrams. The tag is correct.

Method

$$B = \{b, l, u, e\}, \quad G = \{g, r, e, y\}, \quad W = \{w, h, i, t, e\}$$

$B \cup G$ means all the letters that are in B or in G.

$$B \cup G = \{b, l, u, e, g, r, y\}$$

$W \cap G'$ means letters that are in W, but not in G. Since e is in G, it is not included.

$$W \cap G' = \{w, h, i, t\}$$

Serena is not correct because e is in all three sets.

$$B \cap G \cap W = \{e\}$$

FINAL ANSWER

(a)(i) $\{b, l, u, e, g, r, y\}$, (a)(ii) $\{w, h, i, t\}$. Serena is incorrect.

PAST PAPER QUESTION

Q#: 5 Marks: 3

TOPIC: **Area & Perimeter**

Jan 2020 P1HR - Jan 2020 | P1HR

5 The diagram shows Yuen's garden.

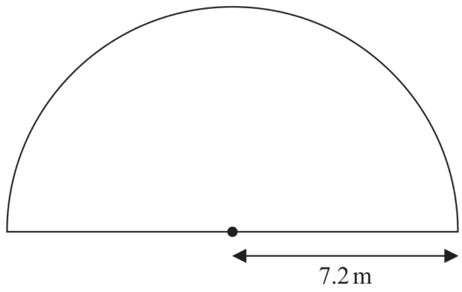

Diagram **NOT** accurately drawn

The garden is in the shape of a semicircle of radius 7.2 m.
Yuen is going to cover his garden with grass seed.

Yuen has 12 boxes of grass seed.
Each box of grass seed contains enough seed to cover 6 m^2 of the garden.

Has Yuen enough grass seed for his garden?
Show your working clearly.

(Total for Question 5 is 3 marks)

PAST PAPER SOLUTION Q#: 5 Marks: 3

TOPIC: **Area & Perimeter**

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Topic check. Area & Perimeter. The tag is correct.

Solution

The garden is a semicircle of radius 7.2 m.

$$\text{area} = \frac{1}{2}\pi r^2$$

$$\text{area} = \frac{1}{2}\pi(7.2)^2 = 81.43\ldots\ \text{m}^2$$

Yuen has 12 boxes, and each box covers 6 m^2:

$$12 \times 6 = 72\ \text{m}^2$$

Since

$$72 < 81.43\ldots$$

Yuen does not have enough grass seed.

FINAL ANSWER
No, he does not have enough grass seed.

PAST PAPER QUESTION

Q#: 6 Marks: 3

TOPIC: **Solving Quadratic Equations**

Jan 2020 P1HR - Jan 2020 | P1HR

6 Solve $x^2 - 5x - 36 = 0$

Show clear algebraic working.

....

(Total for Question 6 is 3 marks)

PAST PAPER SOLUTION

Q#: 6 Marks: 3

TOPIC: **Solving Quadratic Equations**

Jan 2020 P1HR - Jan 2020 | P1HR

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Topic check. Solving quadratic equations. The tag is correct.

Solution

$$x^2 - 5x - 36 = 0$$

Factorise:

$$(x - 9)(x + 4) = 0$$

So

$$x = 9 \quad \text{or} \quad x = -4$$

FINAL ANSWER
$x = 9$ or $x = -4$.

PAST PAPER QUESTION

Q#: 7 **Marks: 3**

TOPIC: **Percentages**

Jan 2020 P1HR - Jan 2020 | P1HR

7 In a sale, the normal price of a hat is reduced by 15%
The sale price of the hat is 20.40 euros.

Work out the normal price of the hat.

.... euros

(Total for Question 7 is 3 marks)

PAST PAPER SOLUTION

Q#: 7 Marks: 3

TOPIC: **Percentages**

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Topic check. Corrected to Percentages. This is a reverse percentage reduction question, not exchange rates.

Method

The hat is reduced by 15%, so the sale price is

$$100\% - 15\% = 85\%$$

of the normal price.

$$0.85 \times \text{normal price} = 20.40$$

$$\text{normal price} = 20.40 \div 0.85 = 24$$

FINAL ANSWER

24 euros.

PAST PAPER QUESTION Q#: 8 Marks: 3

TOPIC: **Statistics Toolkit**

Jan 2020 P1HR - Jan 2020 | P1HR

8 5 children are playing on a trampoline.
The mean weight of the 5 children is 28 kg.

2 of the children get off the trampoline.
The mean weight of these 2 children is 26.5 kg.

Work out the mean weight of the 3 children who remain on the trampoline.

.... kg

(Total for Question 8 is 3 marks)

PAST PAPER SOLUTION

Q#: 8 Marks: 3

TOPIC: **Statistics Toolkit**

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Topic check. Statistics Toolkit. The tag is correct.

Solution

Total weight of 5 children:

$$5 \times 28 = 140 \text{ kg}$$

Total weight of the 2 children who get off:

$$2 \times 26.5 = 53 \text{ kg}$$

Total weight of the 3 children left:

$$140 - 53 = 87 \text{ kg}$$

Mean weight:

$$\frac{87}{3} = 29$$

FINAL ANSWER

29 kg.

PAST PAPER QUESTION

Q#: 9 Marks: 5

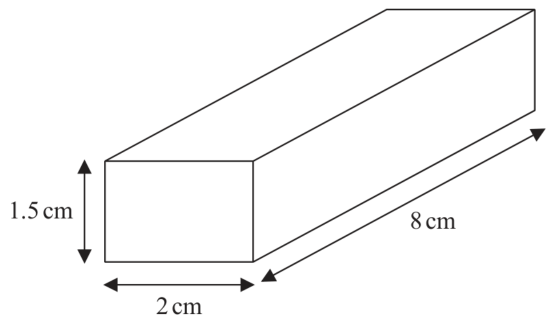
TOPIC: **Standard & Compound Units**

Jan 2020 P1HR - Jan 2020 | P1HR

9 Pablo made a solid gold statue.

He melted down some gold blocks and used the gold to make the statue.
Each block of gold was a cuboid, as shown below.

Diagram **NOT** accurately drawn



The mass of the statue is 5.73 kg.
The density of gold is 19.32 g/cm^3

Work out the least number of gold blocks Pablo melted down in order to make the statue.
Show your working clearly.

....

(Total for Question 9 is 5 marks)

PAST PAPER SOLUTION

Q#: 9 Marks: 5

TOPIC: **Standard & Compound Units**

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Topic check. Correct. This is a density and volume question.

Solution

Convert the mass to grams:

$$5.73 \text{ kg} = 5730 \text{ g}$$

Using

$$\text{density} = \frac{\text{mass}}{\text{volume}}$$

$$\text{volume of gold} = \frac{5730}{19.32} = 296.58\ldots \text{ cm}^3$$

Volume of one cuboid block is

$$1.5 \times 2 \times 8 = 24 \text{ cm}^3$$

Number of blocks needed is

$$\frac{296.58\ldots}{24} = 12.35\ldots$$

Pablo must have a whole number of blocks, so he needs at least 13 blocks.

FINAL ANSWER

13 blocks.

PAST PAPER QUESTION

TOPIC: **Area & Perimeter**

Q#: 10 | Marks: 5

Jan 2020 P1HR - Jan 2020 | P1HR

10 The diagram shows a regular hexagon, *ABCDEF*, and an isosceles triangle, *GHJ*.

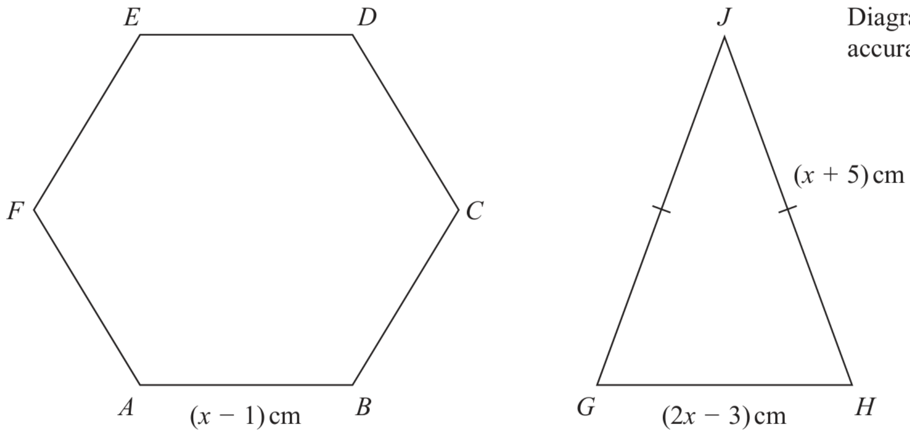

Diagram **NOT** accurately drawn

The perimeter of the hexagon is equal to the perimeter of the triangle.

Find the length of each side of the hexagon.
Show clear algebraic working.

.... cm

(Total for Question 10 is 5 marks)

PAST PAPER SOLUTION

Q#: 10 Marks: 5

TOPIC: **Area & Perimeter**

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WORKED SOLUTION

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Topic check. Corrected to Area & Perimeter. This is a perimeter equation question.

Solution

The regular hexagon has side length $(x-1)$ cm, so its perimeter is

$$6(x-1)$$

The isosceles triangle has perimeter

$$(x+5)+(x+5)+(2x-3)=4x+7$$

The perimeters are equal:

$$6(x-1)=4x+7$$

$$6x-6=4x+7$$

$$2x=13$$

$$x=6.5$$

Each side of the hexagon is

$$x-1=6.5-1=5.5$$

FINAL ANSWER

5.5 cm.

PAST PAPER QUESTION Q#: 11 Marks: 4

TOPIC: **Rounding, Estimation & Bounds**

Jan 2020 P1HR - Jan 2020 | P1HR

11 The weight of a cat is 4.3 kg correct to 2 significant figures.

(a) Write down the upper bound of the weight of the cat.

.... kg

(1)

(b) Write down the lower bound of the weight of the cat.

.... kg

(1)

$G = e - f$

e = 17 correct to the nearest integer

f = 9.4 correct to one decimal place

(c) Work out the upper bound for the value of G.

....

(2)

(Total for Question 11 is 4 marks)

PAST PAPER SOLUTION

Q#: 11 Marks: 4

TOPIC: **Rounding, Estimation & Bounds**

Jan 2020 P1HR - Jan 2020 | P1HR

WORKED SOLUTION

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Topic check. Rounding, Estimation and Bounds. The tag is correct.

Method

The weight 4.3 kg correct to 2 significant figures has bounds

$$4.25 \leq w < 4.35$$

So the upper bound is 4.35 kg and the lower bound is 4.25 kg.
For

$$G = e - f$$

the upper bound uses the upper bound of e and the lower bound of f.

$$e < 17.5, \quad f \geq 9.35$$

$$G_{\text{upper}} = 17.5 - 9.35 = 8.15$$

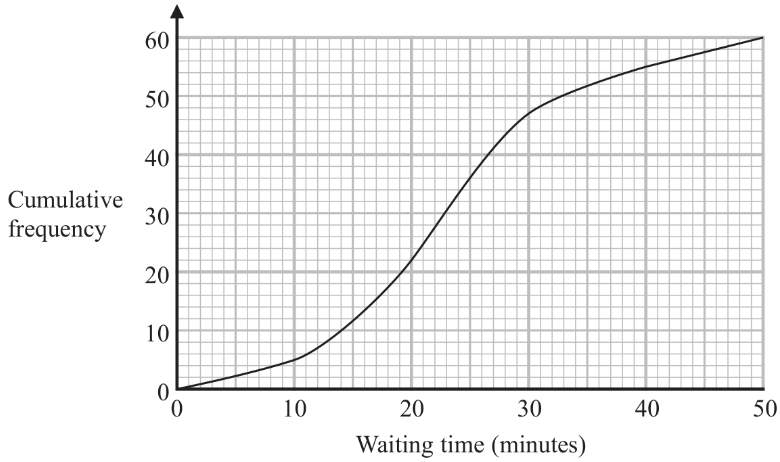
FINAL ANSWER
(a) 4.35 kg, (b) 4.25 kg, (c) 8.15

PAST PAPER QUESTION Q#: 12 Marks: 5

TOPIC: **Cumulative Frequency Diagrams**

Jan 2020 P1HR - Jan 2020 | P1HR

12 The cumulative frequency graph gives information about the waiting times, in minutes, of people with appointments at Hospital A.



(a) Use the graph to find an estimate of the median waiting time at Hospital A.

.... minutes

(1)

(b) Use the graph to find an estimate of the interquartile range of the waiting times at Hospital A.

.... minutes

(2)

At a different hospital, Hospital B, the median waiting time is 28 minutes and the interquartile range of the waiting times is 19 minutes.

(c) Compare the waiting times at Hospital A with the waiting times at Hospital B.

....

....

....

....

(2)

(Total for Question 12 is 5 marks)

PAST PAPER SOLUTION

Q#: 12 Marks: 5

Topic: **Cumulative Frequency Diagrams**

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Topic check. Cumulative Frequency Diagrams. The tag is correct.

Solution

There are 60 people.
The median is at cumulative frequency 30.
From the graph,

$$\text{median} \approx 22.5 \text{ minutes}$$

For the interquartile range:

$$Q_1 \text{ is at } 15, \qquad Q_3 \text{ is at } 45$$

From the graph,

$$Q_1 \approx 16.5, \qquad Q_3 \approx 28.5$$

$$\text{IQR} \approx 28.5 - 16.5 = 12$$

Hospital B has median 28 minutes and IQR 19 minutes.
So Hospital A has a shorter median waiting time and a smaller spread.

FINAL ANSWER

median about 22.5 minutes, IQR about 12 minutes. Hospital A is quicker and more consistent.

PAST PAPER QUESTION

Q#: 13 Marks: 4

TOPIC: **Algebraic Proof**

Jan 2020 P1HR - Jan 2020 | P1HR

13 (a) Use algebra to show that $0.5\dot{7}\dot{2} = \frac{63}{110}$

(2)

Given that y is a prime number,

(b) express $\frac{3}{2-\sqrt{y}}$ in the form $\frac{a+b\sqrt{y}}{c-y}$ where a, b and c are integers.

....

(2)

(Total for Question 13 is 4 marks)

PAST PAPER SOLUTION

Q#: 13 Marks: 4

TOPIC: **Algebraic Proof**

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WORKED SOLUTION

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Topic check. Algebraic proof. The tag is correct.

Solution

(a) Let

$$x = 0.5727272\ldots$$

Then

$$1000x = 572.727272\ldots$$

and

$$10x = 5.727272\ldots$$

Subtract:

$$990x = 567$$

$$x = \frac{567}{990} = \frac{63}{110}$$

So

$$0.5\dot{7}\dot{2} = \frac{63}{110}$$

(b)

$$\frac{3}{2-\sqrt{y}} \times \frac{2+\sqrt{y}}{2+\sqrt{y}} = \frac{3(2+\sqrt{y})}{4-y}$$
$$= \frac{6+3\sqrt{y}}{4-y}$$

FINAL ANSWER

(a) shown, (b) $\frac{6+3\sqrt{y}}{4-y}$

PAST PAPER QUESTION

Q#: 14 Marks: 4

TOPIC: Circle Theorems

Jan 2020 P1HR - Jan 2020 | P1HR

14

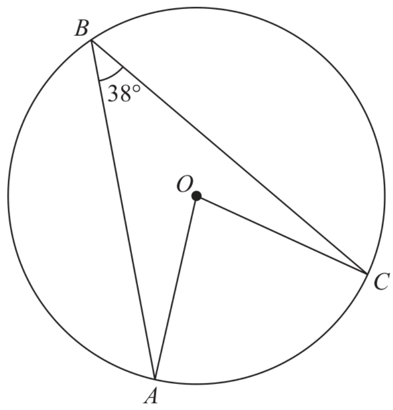

Diagram **NOT** accurately drawn

A, B and C are points on a circle, centre O.
Angle $ABC = 38°$

Work out the size of angle OAC.
Give a reason for each stage of your working.

....°

(Total for Question 14 is 4 marks)

PAST PAPER SOLUTION

Q#: 14 Marks: 4

TOPIC: **Circle Theorems**

Jan 2020 P1HR - Jan 2020 | P1HR

WORKED SOLUTION

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Topic check. Circle Theorems. The tag is correct.

Solution

Angle ABC and central angle AOC stand on the same arc AC.
The angle at the centre is twice the angle at the circumference:

$$\angle AOC = 2 \times 38^\circ = 76^\circ$$

Since $OA = OC$, triangle AOC is isosceles.

$$\angle OAC = \angle OCA = \frac{180^\circ - 76^\circ}{2} = 52^\circ$$

FINAL ANSWER

52°.

PAST PAPER QUESTION

Q#: 15 Marks: 4

TOPIC: **Right-Angled Triangles - Pythagoras & Trigonometry**

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15 The diagram shows two right-angled triangles, *DEF* and *EFG*.

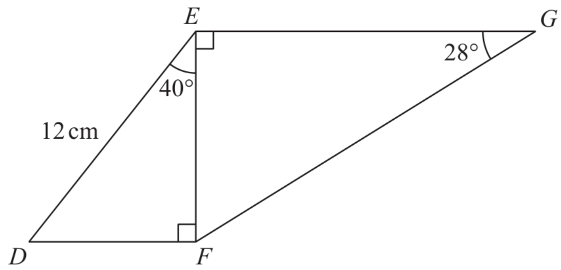

Diagram **NOT** accurately drawn

Work out the length of *EG*.
Give your answer correct to 3 significant figures.

.... cm

(Total for Question 15 is 4 marks)

PAST PAPER SOLUTION

Q#: 15 Marks: 4

TOPIC: **Right-Angled Triangles - Pythagoras & Trigonometry**

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Topic check. Right-Angled Triangles - Pythagoras & Trigonometry. The tag is correct.

Solution

In triangle DEF,

$$EF = 12\cos 40^\circ = 9.192\ldots$$

In triangle EFG,

$$\sin 28^\circ = \frac{EF}{EG}$$

$$EG = \frac{9.192\ldots}{\sin 28^\circ} = 19.580\ldots$$

FINAL ANSWER

19.6 cm.

PAST PAPER QUESTION Q#: 16 Marks: 4

TOPIC: **Combined & Conditional Probability**

Jan 2020 P1HR - Jan 2020 | P1HR

16 Steffi is going to play one game of tennis and one game of chess.

The probability that she will win the game of tennis is 0.6
The probability that she will win **both** games is 0.42

Work out the probability that she will **not** win either game.

....

(Total for Question 16 is 4 marks)

PAST PAPER SOLUTION

Q#: 16 Marks: 4

TOPIC: **Combined & Conditional Probability**

Jan 2020 P1HR - Jan 2020 | P1HR

WORKED SOLUTION

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Topic check. Combined & Conditional Probability. The tag is correct.

Solution

$$P(\text{win tennis}) = 0.6$$

$$P(\text{win both}) = 0.42$$

So the chess win probability used with the tennis result is

$$0.42 \div 0.6 = 0.7$$

Therefore

$$P(\text{lose tennis}) = 0.4, \qquad P(\text{lose chess}) = 0.3$$

$$P(\text{not win either}) = 0.4 \times 0.3 = 0.12$$

FINAL ANSWER

0.12.

PAST PAPER QUESTION Q#: 17 Marks: 4

TOPIC: **Functions**

Jan 2020 P1HR - Jan 2020 | P1HR

17 The function f is such that $f(x) = (x - 4)^2$ for all values of x.

(a) Find f(1)

....

(1)

(b) State the range of the function f.

....

(1)

The function g is such that $g(x) = \dfrac{4}{x + 3}$ $\quad x \neq -3$

(c) Work out fg(2)

....

(2)

(Total for Question 17 is 4 marks)

PAST PAPER SOLUTION

Q#: 17 Marks: 4

TOPIC: **Functions**

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Topic check. Functions. The tag is correct.

Solution

$$f(x) = (x-4)^2$$

(a)

$$f(1) = (1-4)^2 = 9$$

(b) A square is always greater than or equal to zero, so the range is

$$f(x) \geq 0$$

(c)

$$g(x) = \frac{4}{x+3}$$

$$g(2) = \frac{4}{2+3} = \frac{4}{5}$$

FINAL ANSWER

$f(1) = 9$, range $f(x) \geq 0$, and $g(2) = \frac{4}{5}$.

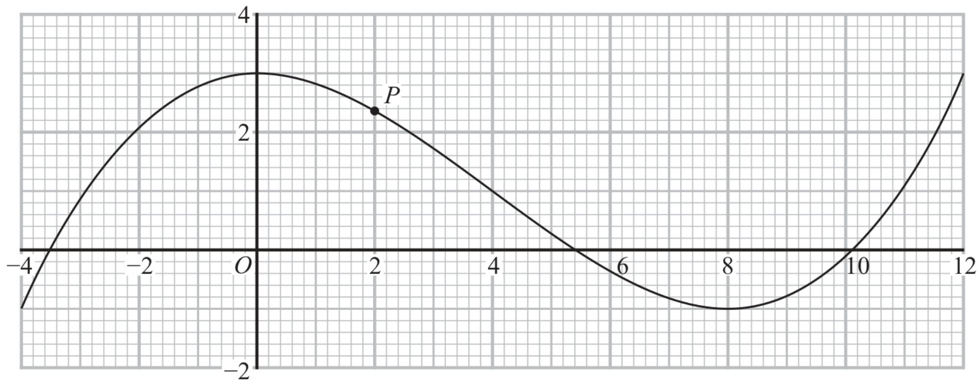
PAST PAPER QUESTION

TOPIC: **Estimating Gradients**

Jan 2020 P1HR - Jan 2020 | P1HR

Q#: 18 Marks: 7

18 The diagram shows the graph of $y = f(x)$ for $-4 \leqslant x \leqslant 12$



The point P on the curve has x coordinate 2

(a) (i) Use the graph to find an estimate for the gradient of the curve at P.

....

(3)

(ii) Hence find an equation of the tangent to the curve at P.
Give your answer in the form $y = mx + c$

....

(2)

The equation $f(x) = k$ has exactly two different solutions for $-4 \leqslant x \leqslant 12$

(b) Use the graph to find the two possible values of k.

.... , ...

(2)

(Total for Question 18 is 7 marks)

PAST PAPER SOLUTION

Q#: 18 Marks: 7

TOPIC: **Estimating Gradients**

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Topic check. Correct. This is estimating gradients from a curve.

Solution

Draw a tangent to the curve at P.
A reasonable tangent gives an estimated gradient of about

$$m \approx -0.6$$

The point P is approximately

$$(2, 2.4)$$

Using $y = mx + c$,

$$2.4 \approx -0.6(2) + c$$

$$c \approx 3.6$$

So an approximate tangent equation is

$$y \approx -0.6x + 3.6$$

The equation $f(x) = k$ has exactly two different solutions when the horizontal line touches one turning level.
From the graph, these levels are approximately

$$k = 3 \quad \text{and} \quad k = -1$$

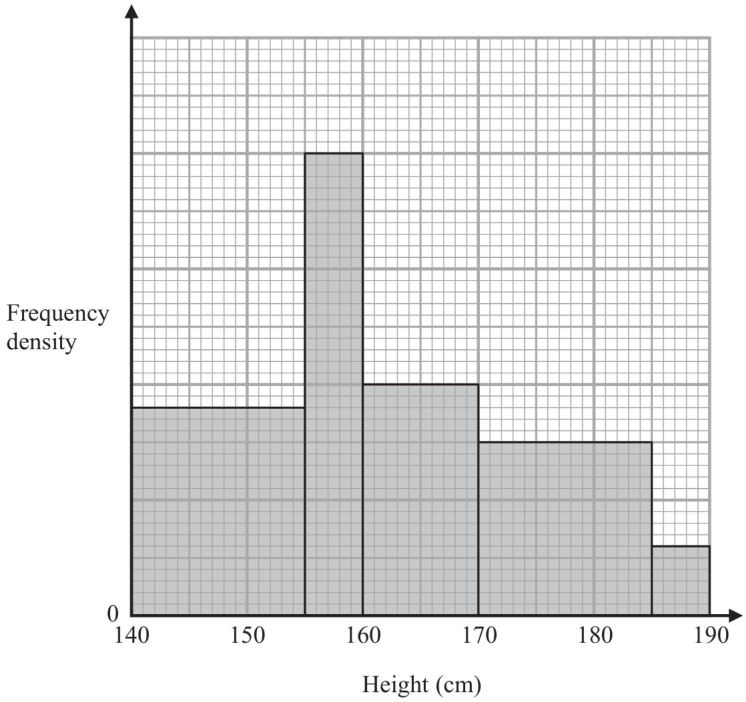
FINAL ANSWER
Gradient ≈ -0.6, tangent $y \approx -0.6x + 3.6$, and $k \approx 3, -1$.

PAST PAPER QUESTION Q#: 19 Marks: 4

Topic: **Histograms**

Jan 2020 P1HR - Jan 2020 | P1HR

19



The histogram gives information about the heights of all the Year 11 students at a school.

There are 160 students in Year 11 with a height between 155 cm and 170 cm.

Work out the total number of students in Year 11 at the school.

....

(Total for Question 19 is 4 marks)

PAST PAPER SOLUTION

Q#: 19 Marks: 4

TOPIC: **Histograms**

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WORKED SOLUTION

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Topic check. Histograms. The tag is correct.

Solution

Use the relative histogram heights.
For $155 < h \leq 170$, the relative area is

$$5(40) + 10(20) = 400$$

This represents 160 students, so each relative area unit represents

$$\frac{160}{400} = 0.4$$

Total relative area:

$$15(18) + 5(40) + 10(20) + 15(15) + 5(6) = 925$$

Estimated total:

$$925 \times 0.4 = 370$$

FINAL ANSWER
370 students.

PAST PAPER QUESTION

Q#: 20 Marks: 5

TOPIC: **Area & Volume of Similar Shapes**

Jan 2020 P1HR - Jan 2020 | P1HR

20 The diagram shows a frustum of a cone and a sphere.

The frustum is made by removing a small cone from a large cone.
The cones are similar.

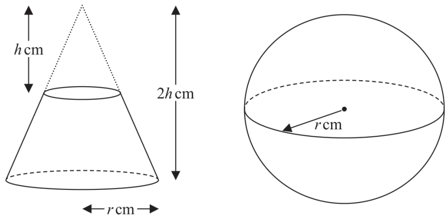

Diagram **NOT** accurately drawn

The height of the small cone is h cm.
The height of the large cone is $2h$ cm.
The radius of the base of the large cone is r cm.

The radius of the sphere is r cm.

Given that the volume of the frustum is equal to the volume of the sphere,

find an expression for r in terms of h.
Give your expression in its simplest form.

r = ...

(Total for Question 20 is 5 marks)

PAST PAPER SOLUTION

Q#: 20 Marks: 5

TOPIC: **Area & Volume of Similar Shapes**

Jan 2020 P1HR - Jan 2020 | P1HR

WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Correct.

Method

The two cones are similar.
The large cone has height $2h$ and base radius r.
The small cone has height h, so its linear scale factor compared with the large cone is

$$\frac{h}{2h} = \frac{1}{2}$$

So the small cone has radius

$$\frac{r}{2}$$

Volume of the large cone:

$$\frac{1}{3}\pi r^2(2h) = \frac{2}{3}\pi r^2 h$$

Volume of the small cone:

$$\frac{1}{3}\pi\left(\frac{r}{2}\right)^2 h = \frac{1}{12}\pi r^2 h$$

So the volume of the frustum is

$$\frac{2}{3}\pi r^2 h - \frac{1}{12}\pi r^2 h = \frac{7}{12}\pi r^2 h$$

The sphere has radius r, so its volume is

$$\frac{4}{3}\pi r^3$$

The two volumes are equal:

$$\frac{7}{12}\pi r^2 h = \frac{4}{3}\pi r^3$$

Divide by πr^2:

$$\frac{7}{12}h = \frac{4}{3}r$$

$$7h = 16r$$

$$r = \frac{7h}{16}$$

FINAL ANSWER

$r = \dfrac{7h}{16}$

PAST PAPER QUESTION

Q#: 20 **Marks: 5**

TOPIC: **Area & Volume of Similar Shapes**

Jan 2020 P1HR - Jan 2020 | P1HR

20 The diagram shows a frustum of a cone and a sphere.

The frustum is made by removing a small cone from a large cone.
The cones are similar.

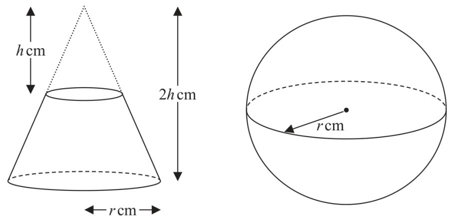

Diagram **NOT** accurately drawn

The height of the small cone is h cm.
The height of the large cone is $2h$ cm.
The radius of the base of the large cone is r cm.

The radius of the sphere is r cm.

Given that the volume of the frustum is equal to the volume of the sphere,

find an expression for r in terms of h.
Give your expression in its simplest form.

r = ...

(Total for Question 20 is 5 marks)

PAST PAPER SOLUTION

Q#: 20 Marks: 5

TOPIC: **Area & Volume of Similar Shapes**

Jan 2020 P1HR - Jan 2020 | P1HR

WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Correct.

Method

The two cones are similar.
The large cone has height $2h$ and base radius r.
The small cone has height h, so its linear scale factor compared with the large cone is

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So the small cone has radius

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$$\frac{1}{3}\pi r^2(2h) = \frac{2}{3}\pi r^2 h$$

Volume of the small cone:

$$\frac{1}{3}\pi\left(\frac{r}{2}\right)^2 h = \frac{1}{12}\pi r^2 h$$

So the volume of the frustum is

$$\frac{2}{3}\pi r^2 h - \frac{1}{12}\pi r^2 h = \frac{7}{12}\pi r^2 h$$

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$$\frac{4}{3}\pi r^3$$

The two volumes are equal:

$$\frac{7}{12}\pi r^2 h = \frac{4}{3}\pi r^3$$

Divide by πr^2:

$$\frac{7}{12}h = \frac{4}{3}r$$

$$7h = 16r$$

$$r = \frac{7h}{16}$$

FINAL ANSWER

$r = \dfrac{7h}{16}$

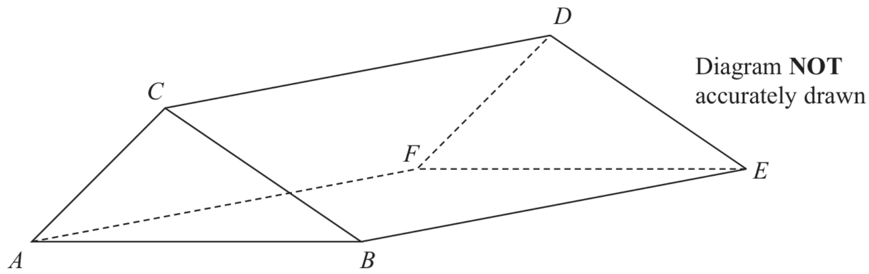
PAST PAPER QUESTION

Q#: 21 **Marks: 6**

TOPIC: **Sine, Cosine Rule & Area of Triangles**

Jan 2020 P1HR - Jan 2020 | P1HR

21 The diagram shows the prism *ABCDEF* with cross section triangle *ABC*.



Angle *BEC* = 40° and angle *ACB* is obtuse.
AC = 6 cm and *CE* = 13 cm

The area of triangle *ABC* is 22 cm²

Calculate the length of *AB*.
Give your answer correct to one decimal place.

.... cm

(Total for Question 21 is 6 marks)

PAST PAPER SOLUTION

Q#: 21 Marks: 6

TOPIC: **Sine, Cosine Rule & Area of Triangles**

Jan 2020 P1HR - Jan 2020 | P1HR

WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Correct.

Method

In triangle BEC, $CE = 13$ and $\angle BEC = 40°$.
Since this is a prism, BE is perpendicular to the cross section, so triangle BEC is right-angled at B.

$$BC = 13 \sin 40°$$

$$BC = 8.356238\ldots$$

The area of triangle ABC is 22 cm^2.
Using

$$\text{Area} = \frac{1}{2}ab \sin C$$

$$22 = \frac{1}{2}(AC)(BC)\sin(\angle ACB)$$

$$22 = \frac{1}{2}(6)(8.356238\ldots)\sin(\angle ACB)$$

$$\sin(\angle ACB) = 0.877587\ldots$$

The angle ACB is obtuse, so

$$\angle ACB = 180° - \sin^{-1}(0.877587\ldots)$$

$$\angle ACB = 118.647\ldots°$$

Use the cosine rule:

$$AB^2 = AC^2 + BC^2 - 2(AC)(BC)\cos(\angle ACB)$$

$$AB^2 = 6^2 + (8.356238\ldots)^2 - 2(6)(8.356238\ldots)\cos(118.647\ldots°)$$

$$AB = 12.4056\ldots$$

Correct to one decimal place:

$$12.4$$

FINAL ANSWER

$\boxed{12.4 \text{ cm}}$

PAST PAPER QUESTION

Q#: 21 Marks: 6

TOPIC: **Sine, Cosine Rule & Area of Triangles**

Jan 2020 P1HR - Jan 2020 | P1HR

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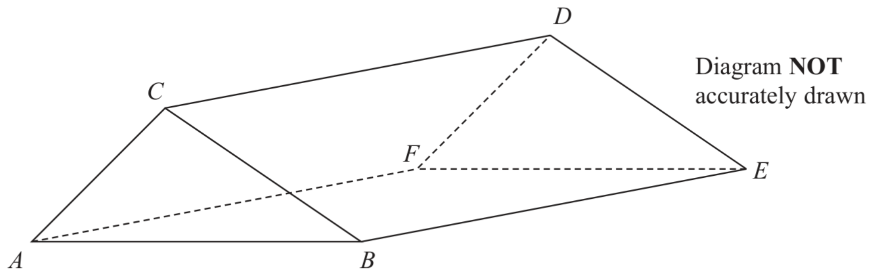

Angle $BEC = 40°$ and angle ACB is obtuse.
$AC = 6$ cm and $CE = 13$ cm

The area of triangle ABC is $22\,\text{cm}^2$

Calculate the length of AB.
Give your answer correct to one decimal place.

.... cm

(Total for Question 21 is 6 marks)

PAST PAPER SOLUTION

Q#: 21 Marks: 6

TOPIC: **Sine, Cosine Rule & Area of Triangles**

Jan 2020 P1HR - Jan 2020 | P1HR

WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Correct.

Method

In triangle BEC, $CE = 13$ and $\angle BEC = 40°$.
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$$\angle ACB = 118.647\ldots°$$

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$$AB^2 = AC^2 + BC^2 - 2(AC)(BC)\cos(\angle ACB)$$

$$AB^2 = 6^2 + (8.356238\ldots)^2 - 2(6)(8.356238\ldots)\cos(118.647\ldots°)$$

$$AB = 12.4056\ldots$$

Correct to one decimal place:

$$12.4$$

FINAL ANSWER

$\boxed{12.4 \text{ cm}}$

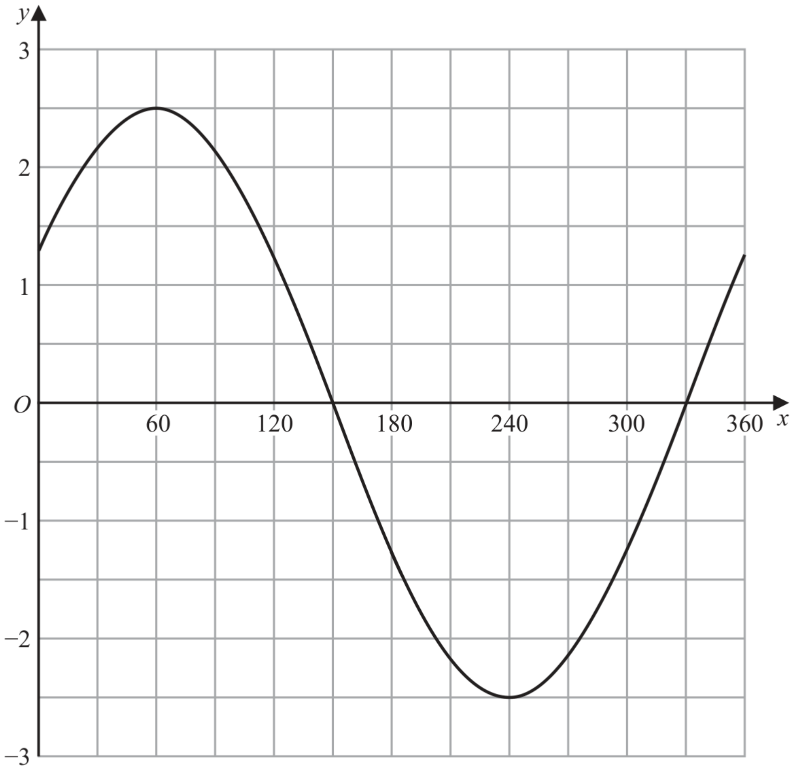
PAST PAPER QUESTION

Q#: 22 Marks: 4

TOPIC: **Transformations of Graphs**

Jan 2020 P1HR - Jan 2020 | P1HR

22 The graph of $y = a\cos(x + b)°$ for $0 \leqslant x \leqslant 360$ is drawn on the grid.



(a) Find the value of a and the value of b.

$a =$ ....

$b =$ ....

(2)

Another curve C has equation $y = f(x)$
The coordinates of the minimum point of C are (4, 5)

(b) Write down the coordinates of the minimum point of the curve with equation

(i) $y = f(2x)$

(.................... , ...)

(ii) $y = f(x) - 7$

(.................... , ...)

(2)

(Total for Question 22 is 4 marks)

PAST PAPER SOLUTION

Q#: 22 Marks: 4

TOPIC: **Transformations of Graphs**

Jan 2020 P1HR - Jan 2020 | P1HR

WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Correct.

Method

For part (a), the graph has maximum 2.5 and minimum -2.5, so

$$a = 2.5$$

The maximum occurs at $x = 60°$. For $y = a\cos(x + b)°$, a maximum occurs when

$$x + b = 360$$

So

$$60 + b = 360$$

$$b = 300$$

For part (b), the minimum point of $y = f(x)$ is $(4, 5)$.
For

$$y = f(2x)$$

the x-coordinate is divided by 2:

$$(4, 5) \to (2, 5)$$

For

$$y = f(x) - 7$$

the y-coordinate decreases by 7:

$$(4, 5) \to (4, -2)$$

FINAL ANSWER

$a = 2.5,\ b = 300;\ (2, 5)$ and $(4, -2)$

PAST PAPER QUESTION

Q#: 22 Marks: 4

TOPIC: **Transformations of Graphs**

Jan 2020 P1HR - Jan 2020 | P1HR

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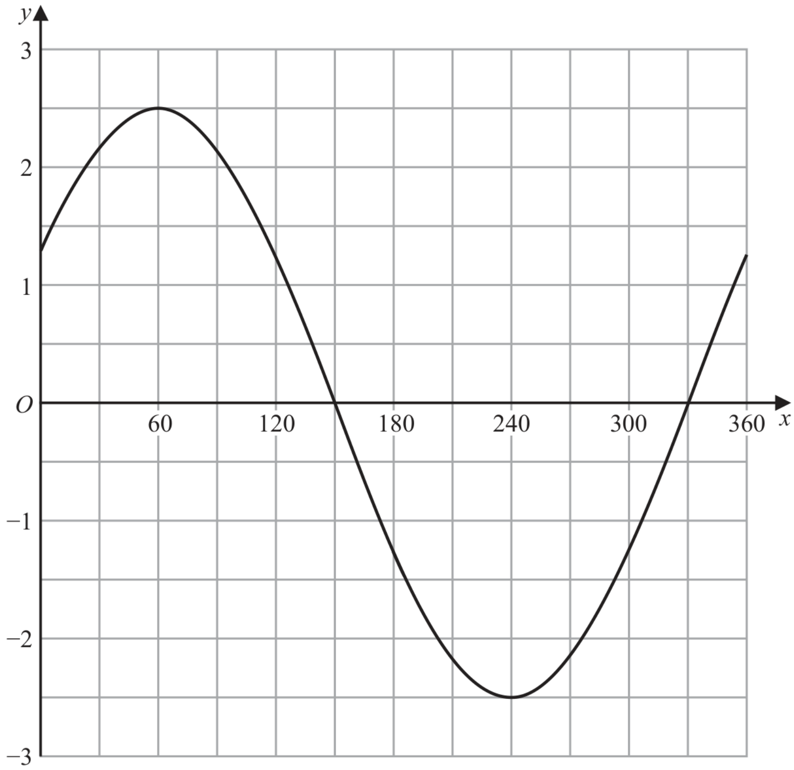

(a) Find the value of a and the value of b.

a = ...

b = ...

(2)

Another curve C has equation $y = \mathrm{f}(x)$
The coordinates of the minimum point of C are (4, 5)

(b) Write down the coordinates of the minimum point of the curve with equation

(i) $y = \mathrm{f}(2x)$

(.................... , ...)

(ii) $y = \mathrm{f}(x) - 7$

(.................... , ...)

(2)

(Total for Question 22 is 4 marks)

PAST PAPER SOLUTION

Q#: 22 Marks: 4

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Jan 2020 P1HR - Jan 2020 | P1HR

WORKED SOLUTION

Dr. Eslam Ahmed

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$$(4, 5) \to (2, 5)$$

For

$$y = f(x) - 7$$

the y-coordinate decreases by 7:

$$(4, 5) \to (4, -2)$$

FINAL ANSWER

$a = 2.5,\ b = 300;\ (2, 5)$ and $(4, -2)$

PAST PAPER QUESTION

Q#: 23 Marks: 6

TOPIC: **Differentiation**

Jan 2020 P1HR - Jan 2020 | P1HR

23 A particle moves along a straight line.
The fixed point O lies on this line.
The displacement of the particle from O at time t seconds, $t \geqslant 0$, is s metres where

$$s = t^3 + 4t^2 - 5t + 7$$

At time T seconds the velocity of P is V m/s where $V \geqslant -5$

Find an expression for T in terms of V.

Give your expression in the form $\dfrac{-4 + \sqrt{k + mV}}{3}$ where k and m are integers to be found.

$T =$ ....

(Total for Question 23 is 6 marks)

PAST PAPER SOLUTION

Q#: 23 Marks: 6

TOPIC: **Differentiation**

Jan 2020 P1HR - Jan 2020 | P1HR

WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Correct.

Method

$$s = t^3 + 4t^2 - 5t + 7$$

Velocity is the derivative of displacement:

$$v = \frac{ds}{dt} = 3t^2 + 8t - 5$$

At time T, the velocity is V, so

$$V = 3T^2 + 8T - 5$$

$$3T^2 + 8T - (V + 5) = 0$$

Use the quadratic formula:

$$T = \frac{-8 \pm \sqrt{8^2 - 4(3)(-(V+5))}}{2(3)}$$

$$T = \frac{-8 \pm \sqrt{64 + 12V + 60}}{6}$$

$$T = \frac{-8 \pm \sqrt{124 + 12V}}{6}$$

$$T = \frac{-4 \pm \sqrt{31 + 3V}}{3}$$

Since $T \geq 0$, take the positive root:

$$T = \frac{-4 + \sqrt{31 + 3V}}{3}$$

FINAL ANSWER

$$T = \frac{-4 + \sqrt{31 + 3V}}{3}$$

PAST PAPER QUESTION

Q#: 23 **Marks: 6**

TOPIC: **Differentiation**

Jan 2020 P1HR - Jan 2020 | P1HR

23 A particle moves along a straight line.
The fixed point O lies on this line.
The displacement of the particle from O at time t seconds, $t \geqslant 0$, is s metres where

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At time T seconds the velocity of P is V m/s where $V \geqslant -5$

Find an expression for T in terms of V.

Give your expression in the form $\dfrac{-4 + \sqrt{k + mV}}{3}$ where k and m are integers to be found.

$T =$ ....

(Total for Question 23 is 6 marks)

PAST PAPER SOLUTION

Q#: 23 Marks: 6

TOPIC: **Differentiation**

Jan 2020 P1HR - Jan 2020 | P1HR

WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Correct.

Method

$$s = t^3 + 4t^2 - 5t + 7$$

Velocity is the derivative of displacement:

$$v = \frac{ds}{dt} = 3t^2 + 8t - 5$$

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$$T = \frac{-8 \pm \sqrt{124 + 12V}}{6}$$

$$T = \frac{-4 \pm \sqrt{31 + 3V}}{3}$$

Since $T \geq 0$, take the positive root:

$$T = \frac{-4 + \sqrt{31 + 3V}}{3}$$

FINAL ANSWER

$$T = \frac{-4 + \sqrt{31 + 3V}}{3}$$